

# How the EU Soil Observatory is providing solid science for healthy soils

## (Data Documentation - Metadata)

### 1. Geotiffs

#### **multiband.tif**

**Dataset:** Soil degradation indicators in EU

**Extent:** 2636073,1385914 : 6526073,5421914

**Coordinate Reference System (CRS):** EPSG:3035 - ETRS89-extended / LAEA Europe

**Coordinates Units:** meters

**Data type:** Byte - Eight bit unsigned integer

**Pixel Size:** 500

**Number of bands:** 20

**Bands Description:**

**Band 1:** Arsenic

**Band 2:** Cadmium

**Band 3:** Copper

**Band 4:** Mercury

**Band 5:** Nitrogen surplus

**Band 6:** Phosphorus deficiency

**Band 7:** Phosphorus excess

**Band 8:** Soil compaction (Packing Density)

**Band 9:** Peatland degradation risk

**Band 10:** Post-fire recovery

**Band 11:** Harvest erosion

**Band 12:** Secondary salinization risk

**Band 13:** Potential threat to biological functions

**Band 14:** Distance to maximum SOC level

**Band 15:** Soil sealing

**Band 16:** Tillage Erosion

**Band 17:** Water erosion

**Band 18:** Wind erosion

**Band 19:** Zinc

**Band 20:** SUM of Indicators

**Pixel Values for Bands 0-19:**

**0** = Healthy

**1**= Unhealthy

**Nodata value for all bands= 127.**

**Pixel Values for Band 20:**

**0** = Healthy

**1-8**= Number of concurrent indicators

**126**= Soil Sealing

## 2. Cite

Panagos, P., Broothaerts, N., Ballabio, C., Orgiazzi, A., De Rosa, D., Borrelli, P., Liakos, L., Vieira, D., Van Eynde, E., Arias Navarro, C., Breure, T., Fendrich, A., Köninger, J., Labouyrie, M., Matthews, F., Muntwyler, A., Jimenez, J.M., Wojda, P., Yunta, F., Marechal, A., Sala, S., Jones, A., 2024. How the EU Soil Observatory is providing solid science for healthy soils. European J Soil Science 75, e13507. <https://doi.org/10.1111/ejss.13507>